Graphic Era HILL UNIVERSITY
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University under section 2(f) of UGC Act, 1956

# Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: **BTech (ECE) III**
Name of the Paper: **Signal & System**
Paper Code: **TEC 304**
Time: 1.5 hour

Maximum Marks: 50

**Note:**

*(i)* Answer all the questions by choosing any one of the sub questions
*(ii)* Each question carries 10 marks.
*(iii)* Please specify COs against each question.

Q1. (10 Marks)

a. What basic operations can be performed on continuous-time signals? Give examples of how these operations change the signals. CO1

OR

b. A signal x(t)=3t+2 is given. Check if the signal is even, odd, or neither. CO1

Q2. (10 Marks)

a. How do even and odd signals differ from each other? How can you tell if a signal is even, odd, or neither? Provide examples for each. CO1

OR

b. Determine whether the following signals x(t)=sin(2t) + cos(2t) & x(t)=0.5 + cos(t) are periodic. If so, find their fundamental period. CO1

Q3. (10 Marks)

a. Define energy and power signals. Provide the conditions for a signal to be classified as an energy signal or a power signal. Also explain what are deterministic and random signals? Give examples of each. CO1

OR

b. For a continuous-time signal $x(t)=e^{-2t}u(t)$ & x(t)=A calculate whether the signal is an energy signal or a power signal. CO1

Q4. (10 Marks)

a. Discuss the key differences between linear and nonlinear systems, and between time-invariant and time-variant systems. Give the examples also. CO2

OR

b. Given an LTI system with impulse response $h(t)=e^{-t}u(t)$, find the output of the system when the input is x(t)=u(t) (the unit step function). CO2

Q5. (10 Marks)

a. How can you classify a system as memoryless, causal, or time-invariant by analyzing its system equation or response? Provide examples for each classification. CO2

OR

b. Determine whether the system $y(t)=x^2(t)$ is static or dynamic, linear or nonlinear, causal or non-causal, and time-invariant or time-variant. CO2